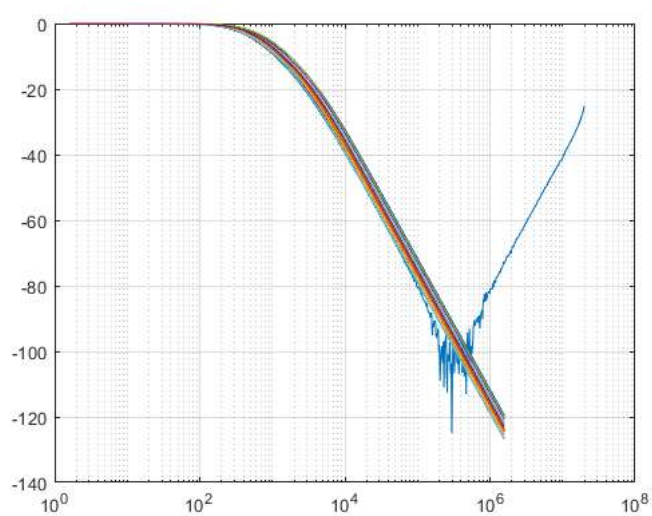
```
clear all;
close all;
clc

A=dlmread('scope_1.2_modified.csv');
x = A(:,1);
y = A(:,2);
semilogx(x,y)
grid on
hold on
QX = [66.67, 250, 1*10^-6, 1*10^-6];
b =[1;
    0;
    0];
w=0:10:10e6;
F = freqresp5(QX,b,w,20);
semilogx(w/(2*pi),F)
grid on
xscale log
hold off
function F = freqresp5(QX,b,w,num)
F=[];
% QX is a vector that contains all the nominal element values. % Resistors are listed first (as Q(1),Q(2) and Q(3)), then % capacitors (Q(4)), and finally inductors
for s=1:num
    Q=variation(QX);
    % This function produces random variations of the element % values, which are within 20% of the nominal ones.
    G=zeros(3,3); C=zeros(3,3); L=zeros(3,3);
    % This line saves us some time, since we now need to enter % only the nonzero elements in G, C and L.
    G(1,1)=1;G(2,1)=-1/Q(1);G(2,2)=1/Q(1)+1/Q(2);G(2,3)=-1/Q(2);
    G(3,2) = -1/Q(2); G(3,3)=1/Q(2);
    C(2,2)= Q(3); C(3,3)=Q(4);
    % For each new set of element values, matrices G, C and L must % be recomputed.
    mag=zeros(3,1);
    % Vector mag is initialized.
    for k=1:length(w)
        omega=w(k);
        % Omega is the next frequency in w.
        A=G+i*omega*C;
        x=A\b;
        % Vector x contains the solution of the node voltage % equations in complex form.
        mag= [mag abs(x)];
        % The magnitudes of all voltages as stored as a new column in % matrix mag.
    end
        V3=mag(3,2:length(w)+1);
        %Since we are interested only in V3, we will extract row 3 of matrix % mag (and ignore the first column).
        FX=20*log10(V3);
        hold on;
        F=[F;FX];
end
end
function Q=variation (R)
    m=length(R);
    g=ones(1,m)+.5*rand(1,m);
    Q=.8*(R.*g);
end
```



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